

MD ASADULLAHIL GALIB FARDIN

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Dhaka, Bangladesh

ACADEMIC CREDENTIALS

MSc in Climate Modelling and Risk Management
Bangladesh University of Engineering and Technology (BUET)

July 2026 (Tentative)
CGPA: 3.33/4.00 (ECE Evaluated)
CGPA: 3.17/4.00 (Before Evaluation)

Relevant Coursework:

Numerical Modelling in Water and Sediment Transport, Hazards and Risk Management, Applications of Remote Sensing in Climate Change

BSc in Civil Engineering
Rajshahi University of Engineering and Technology (RUET)

August 2024
CGPA: 3.35/4.00 (ECE Evaluated)
CGPA: 3.31/4.00 (Before Evaluation)

Relevant Coursework:

Fluid Mechanics, Hydrology, Engineering Hydraulics, Irrigation and Flood Engineering

STANDARDIZED SCORES

IELTS Academic Test: 8.0

Listening: 9.0 Reading: 8.5 Writing: 7.5 Speaking: 6.5

4th November, 2025

ACADEMIC RESEARCH EXPERIENCE

Undergraduate Thesis: *"Dynamic Analysis of Rooppur Nuclear Power Plant Under Seismic and Wind Loading"*

1. Developed a Finite Element model of the Cooling tower in the **ABAQUS** FEA software.
2. The shell structure of the tower was modelled as the Continuum shell and the Conventional shell element
3. Two types of column modelling approaches were taken, including the simplified approach and the heterogeneous design approach and the performance of each modelling was observed during loading
4. Non-linear time history analysis due to earthquake loading was conducted
5. Wind loading was calculated using the Greiner equation approach for cylindrical shell and was applied using the Analytical Field option available in the ABAQUS software.

Selected MSc Term Projects:

1. Numerical and Analytical Solutions of the One-Dimensional Advection–Dispersion Using the FTCS Scheme
GitHub Repository
2. Hydrodynamic Modelling of the Halda River through Explicit and Semi-Implicit Schemes for Water and Sediment Transport Analysis
GitHub Repository
3. Hydrodynamic Assessment of Tidal-Induced Backwater and Bridge-Induced Velocity Effects on The Halda River Using HEC-RAS
GitHub Repository

4. Investigation of Remote Sensing-Based Agricultural Drought Monitoring in Bangladesh
GitHub Repository
5. Evaluation of Reanalysis and Satellite-Based Precipitation Products Across Bangladesh
GitHub Repository
6. Bias Correction of CORDEX 1-day Precipitation Data, Rainfall Trend Analysis and Drought Prediction
GitHub Repository

PUBLICATIONS

Journal Articles

1. Chy, I.U.A., **Fardin, M.A.G.**, Awall, M.R. et al. (2025). *Dynamic analysis of cooling tower considering two different column modeling approaches*. **Journal of Building Rehabilitation**, 10, 120. DOI: 10.1007/s41024-025-00632-6. **Published**
2. Rony, K.R.I., **Fardin, M.A.G.**, Chy, I.U.A. et al. (2025). *Modelling Of CFRP Strengthened Beam-Column Connections Under Monotonic and Cyclic Loading*. **Advances in Civil Engineering**. **Under Review**
3. Afroz, Rownok, **Fardin, M.A.G.** et al. (2026). *Impact Behaviour OF Pressurised uPVC Pipe: An Experimental and Numerical Analysis*. **Advances in Civil Engineering** **Under Review**

Conference Papers

1. Ahmed Dipu, S. U., Gemintang, E., Haque Laskor, Md. A., Bhuiyan, F., **Galib Fardin, M. A.**, Rahman, Md. M., Rabbany, Y., and Islam Fahim, S. *Hydrodynamic Changes of Estuarine Islands in the Meghna River under Future Climate Scenarios*. EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-17766, 2026
2. Jahib, A. A., **Fardin, M. A. G.**, and Khan, S. M.: Post-Flood Sediment Deposition Analysis Using SAR Interferometry: A Case Study of the 2024 Flood in Feni district, Bangladesh. EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-13958, 2026.

TECHNICAL SKILLS

1. **Modeling Tools:** ABAQUS, ETABS, Delft3D, OpenSees
2. **Geospatial:** SNAP, Google Earth Engine, ArcGIS Pro
3. **Programming:** Python, MATLAB
4. **Drafting & Design:** AutoCAD, SketchUp